

ECDLP hardness across the ECC2K-130 isogeny class

E0, all 262 conductor-263 floor curves, and the unenumerated conductor-p and conductor-263p levels: a per-curve validation with adversarial verification

Answer. No curve in the class is easier than E0. The best known ECDLP cost is $2^{60.81}$ **ideal Pollard-rho iterations for E0 and for every one of the 262 floor curves** ($2^{60.91}$ with the Bailey et al. 2009 walk). A floor curve is intrinsically $\sqrt{131} \approx 11.4\times$ (3.52 bits) harder for rho because it has no Frobenius self-map, but its unique rational 263-isogeny up to E0 was built and verified on all 262 curves and costs at most $2^{15.6}$ field multiplications, which removes the gap. The only levels where hardness could differ are conductor p and 263p ($p = 146505763881528721 \approx 2^{57}$); there it lies between $2^{60.81}$ and $2^{64.33}$, and no such curve can be constructed at feasible cost.

Level (conductor)	End	# curves	Explicitly known	Native rho (log2)	Effective (log2)	Why
1 (crater)	$O_K = Z[\tau]$	1	E0	60.81	60.81	rho on $(-1, \tau)$ classes of size 262
263 (floor)	$Z + 263 \cdot O_K$	262	all 262	64.33	60.81	unique rational 263-isogeny to E0, verified 262/262; $\leq 2^{15.6} F_q$ -mults per instance
p	$Z + p \cdot O_K$	$p + 1 \approx 2^{57.02}$	none	64.33	[60.81, 64.33]	lower end needs an unimplemented characteristic-2 Kani (higher-dimensional) isogeny pipeline
263p (= $Z[\pi]$)	$Z[\pi]$	$262(p + 1) \approx 2^{65.06}$	none	64.33	[60.81, 64.33]	cheap 263-step, then as level p

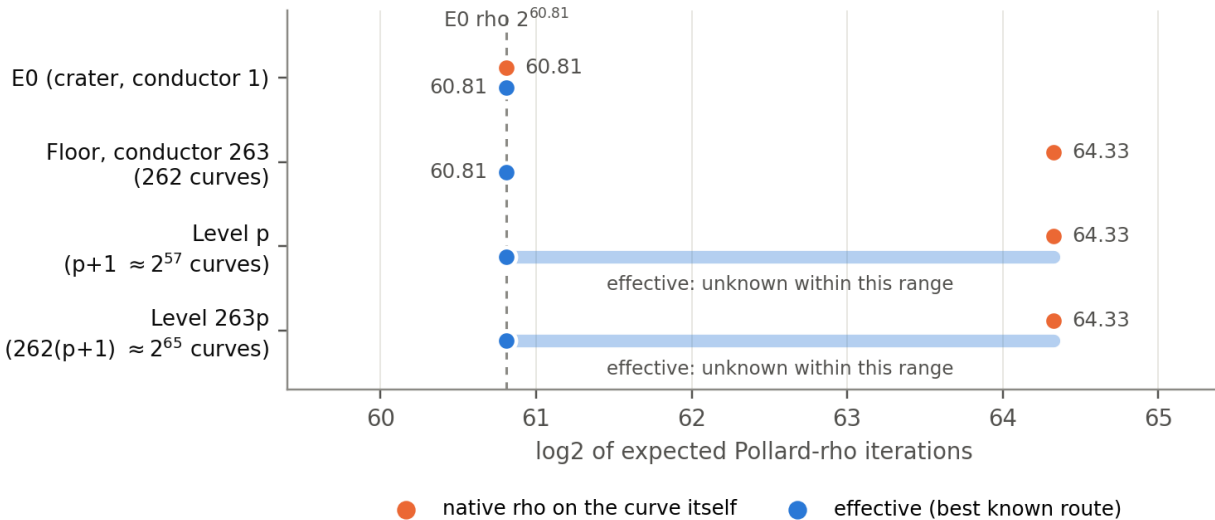


Figure 1. Expected rho iterations by volcano level. Orange: rho run on the curve itself. Blue: best known route (for the floor, transport to E0 first). For the p-levels the effective cost is only known to lie in the shaded interval.

1. Scope and method

Target: every curve F -isogenous to the Certicom ECC2K-130 Koblitz curve $E0 : y^2 + xy = x^3 + 1$ over F_q , $q = 2^{131}$ (polynomial basis modulo $z^{131} + z^{13} + z^2 + z + 1$). Isogenous curves share the group order $4N$, so the question is whether the extra structure of each curve (endomorphism ring, coefficient b , twist, torsion) or its position in the isogeny volcano changes the cost of the discrete logarithm in the order- N subgroup.

The analysis ran as two multi-agent workflows: 184 agents in total. Two agents built the ground truth independently. Twelve attack-vector sweeps each covered all 263 explicit curves, and every claim was checked by two further verifiers, one recomputing with independent code and one auditing the original scripts. Three red-team agents looked for curve-specific weaknesses, a completeness critic ran six gap-fill tasks, and a synthesis agent merged the per-curve table. 129 agents completed; 55 were lost to account usage limits and backfilled as described in §9. No attempt was made on the real challenge DLP; every solved DLP is a toy instance with a planted logarithm.

Constants (all re-derived): trace $t = -22283658519494248867$; $\#E = 4N$ with $N = 680564733841876926932320129493409985129$ (prime, Pocklington and Morrison proofs, $\log_2 N = 129.0$); $t^2 - 4q = -7f^2$ with $f = 263 \cdot p$, $p = 146505763881528721$ prime and inert in $Q(\sqrt{-7})$, 263 split. Class numbers of the four orders: $h = 1, 262, p + 1, 262(p + 1)$ for conductors 1, 263, p , $263p$.

2. The isogeny class and its ground truth

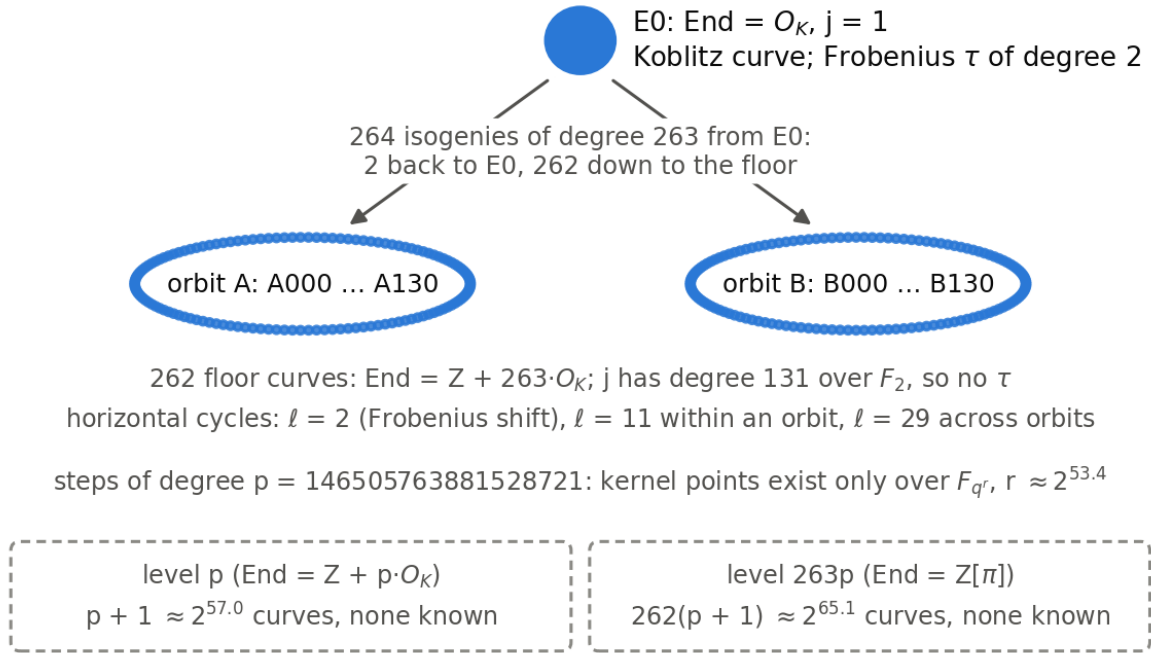


Figure 2. The $\ell = 263$ volcano. $E0$ alone on the crater; 262 floor curves in two Frobenius orbits of 131. Levels p and $263p$ hang below through degree- p isogenies and are not enumerated.

- **Build.** Ring class polynomial $H_{-7, 263}$ (PARI polclass) is squarefree mod 2 and splits into two irreducible factors of degree 131, giving 262 roots in F_q . All 263 curves were point-counted in both twists; $a_2 = 0$ is the order- $4N$ twist on all 263, confirmed by point-order proofs.
- **Independent derivation.** Vélu isogenies from a basis of $E0[263]$ over F_{q^2} , using neither H_D nor Φ_{263} : **262/262 j -invariants match, none extra on either side.** PARI polmodular confirms $\Phi_{263}(1, Y) \equiv (Y + 1)^2 \cdot H_D(Y) \pmod{2}$.
- **Labels.** Codex's labels (A000-A130, B000-B130, X_k with $j = j(X000)^{2^k}$) are reproduced exactly, confirmed four ways including the H_D coefficient hash.

- **Structural point.** Coordinate Frobenius $(x, y) \mapsto (x^{2^k}, y^{2^k})$ is a free group isomorphism $A_{000} \rightarrow A_k$, so up to relabelling the 262 floor curves are two curves.

3. Attack vectors, curve by curve

Every row covers all 263 explicit curves. “Verification” is the outcome of the independent recompute and audit.

Vector	Result on all 263 explicit curves	Effect	Verification
Group order / Pohlig-Hellman	#E = 4N on 263/263; $E(F_q) \cong Z/4N$ cyclic, 2-part $Z/4$ (never $Z/2 \times Z/2$)	0 bits (N prime)	confirmed $\times 2$ (C1-C3); C4 recompute confirmed
Endomorphism ring	$\text{End}(E_0) = O_{K'}; \text{End} = O_{263}$ on 262/262 by ≥ 3 criteria (H_D root, Vélú to $j = 1$, Φ_{263} root structure, 263-Sylow of $E(F_{q^2})$: cyclic $Z/263^2$ on the floor, $(Z/263)^2$ on E_0)	defines the levels	confirmed $\times 2$ (C1-C3, C5); C4 by gap G1
MOV / Frey-Rück	embedding degree $k = \text{ord}_N(q) = 216464610000596986937760855436835237$ (118 bits), identical on every curve by Tate's theorem	0 bits	confirmed $\times 2$
Anomalous / supersingular	not applicable anywhere in the class ($\text{char } 2, N \neq 2$, ordinary)	0 bits	confirmed $\times 2$
Pairings on 263-torsion	$t_{263}(P, Q_N) = 1$ on every curve: blind to the N-part	0 bits	confirmed (recompute + gap G6)
GHS / Weil descent	magic number $m = 1$ on E_0 (the cover is E_0 over F_2 , order 4, kills the N-part); $m = 131$ on all 262 floor curves (genus 2^{130}). 131 is prime, so F_2 is the only subfield	0 bits	re-verified on all 263 (gap G2)
Rho automorphisms / endomorphisms	$\text{Aut} = \{\pm 1\}$; E_0 adds τ (eigenvalue order 131 mod N). On the floor any endomorphism with eigenvalue order $< 2^{20}$ (swept exactly to 2^{32}) has degree $\geq 2^{114.6}$; Codex's $\psi = (11\text{-isogeny}) \circ \text{Frob}^{16}$ has eigenvalue order $(N - 1)/3$	E_0 class 262; floor class 2	re-verified with independent code (gap G1)
Isogeny transport (floor $\rightarrow E_0$)	unique rational 263-isogeny built on 262/262; codomain $j = 1$; planted-scalar homomorphism checks pass; dual composition = $-[263]$; three implementations agree	floor effective = 60.81	9 confirmed, 1 partially (cost wording)
Index calculus (summation polynomials)	rational-x densities indistinguishable from random curves (Figure 4); no method family beats rho, the closest non-generic model is ≥ 23.6 bits slower; Codex's E_0 d_{reg} advantage is a basis-presentation effect (0 bits after minimising over presentations)	0 bits	7 confirmed, 3 partially; gaps G3, G5
Twist / x-only oracle	leaks $\pm k \bmod 526$ on E_0 (8.04 bits), $\pm k \bmod 2 \cdot 263^2$ on floor curves (16.08 bits); twist rho $2^{53.27}$ (E_0 's twist, $\pm \tau$) and $2^{56.79}$ (floor twists)	unvalidated implementations only; ECDLP unchanged	partially confirmed (corrected)
Batch / multi-instance	L pooled instances: amortised gain $0.5 \cdot \log_2 L - 0.675$ bits (3.35 bits at $L = 263$)	no single DLP below $2^{60.81}$	partially confirmed

4. Toy-scale end-to-end evidence

Koblitz analogues over F_{2^n} were chosen so that a small split prime ℓ divides the Frobenius conductor, giving the same volcano shape. For each, DLPs with planted logarithms were solved three ways: rho on a floor curve (negation only), rho on E_0 (negation and τ classes), and transport floor $\rightarrow E_0$ followed by E_0 rho. All 24,300 distinct planted DLPs (52,600 solver runs) were recovered and checked against the planted scalar. Two independent implementations were used.

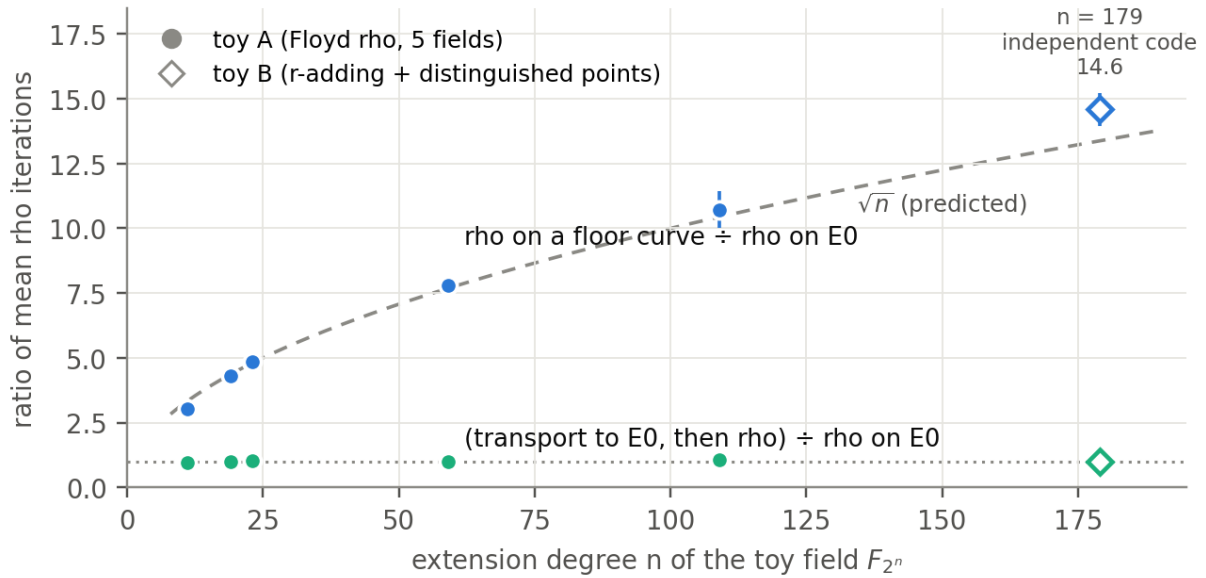


Figure 3. Ratios of mean rho iterations with 95% intervals. Floor/E0 tracks $\sqrt{n}$, the size of E0's Frobenius equivalence classes; transport-then-E0 matches E0 (ratio 1). Toy B's excess at $n = 179$ comes from fruitless-cycle handling on the floor walk.

Toy	n	ℓ	N (bits)	floor / E0 ($\pm$ s.e.)	$\sqrt{n}$	(transport + E0) / E0
A	11	23	9.9	3.03 ± 0.08	3.32	0.962 ± 0.024
A	19	457	18.0	4.29 ± 0.06	4.36	0.998 ± 0.013
A	23	967	21.0	4.84 ± 0.07	4.80	1.021 ± 0.016
A	59	5783	33.2	7.81 ± 0.12	7.68	1.003 ± 0.017
A	109	3271	44.8	10.72 ± 0.36	10.44	1.045 ± 0.034
B	179	359	35.8	14.59 ± 0.33	13.38	0.980 ± 0.022

A second toy level with a larger conductor prime (1721) mirrors the p-level situation: there, transport cost about 4,800 $\times$ the rho cost on the level curve itself, so the level curve was effectively as hard as its native rho.

5. The conductor-p and conductor-263p levels

- Frobenius acts on $E_0[p]$ as the scalar $c = t/2 \bmod p = 65861677189822723$, of multiplicative order $r = 12208813656794060 \approx 2^{53.44} = (p - 1)/12$. Points of order p therefore exist only over F_q ; their x-coordinates lie in $F_{q^{r/2}}$, and one such field element is about $2^{59.5}$ bits.
- Classical routes are infeasible: kernel points $\geq 2^{111.9} F_q$ -mults; the modular polynomial $\Phi_p(\tilde{O}(p^2)) \approx 2^{114}$; complex multiplication with $|D| = 7p^2 \approx 2^{116.9}$.
- Galbraith, “Climbing and descending tall isogeny volcanos”** (Research in Number Theory 11, 2025; ePrint 2024/924): Theorem 1 computes an evaluable representation of an unknown N -isogeny between two *given* curves in $\tilde{O}(N^{1/2})$, using Kani/Robert embeddings of dimension 4, 6 or 8. For $N = p$ that climbs a presented level- p curve to E_0 in $\tilde{O}(2^{28.5})$. But the theorem as proved uses dimension 8, which models at $\geq 2^{72} F_q$ -mults: worse than native rho, whose break-even point is $\approx 2^{66.5} - 2^{67.2}$.
- Dimension-2 (not in the paper) and dimension-4 (heuristic) variants model at about $2^{52.5} - 2^{55.5} \cdot \kappa F_q$ -mults. Every published implementation of such isogeny chains assumes odd characteristic, and Theorem 1 uses $E[4]$ and level-2 theta structures that do not carry over to characteristic 2 directly. A characteristic-2 Kani pipeline (gluing and splitting) does not exist.
- Hence the effective hardness is in [60.81, 64.33].** If the characteristic-2 machinery were built with modest overhead, it would be about 60.9–62.0.

- **Construction.** There is no structural shortcut. Generic search needs $\approx 2^{64.94}$ candidates per hit; with free 2-adic prefilters (#E mod 16, 32 and 64 read off low coefficients of b's characteristic polynomial), the best known cost is $\approx 2^{70.0} - 2^{71.3}$ F_q -mults, about $2^7 - 2^8$ E0 rho solves. That is more than solving natively on the curve once found, so no ECDLP instance can feasibly sit on these levels.

6. Index calculus across the class

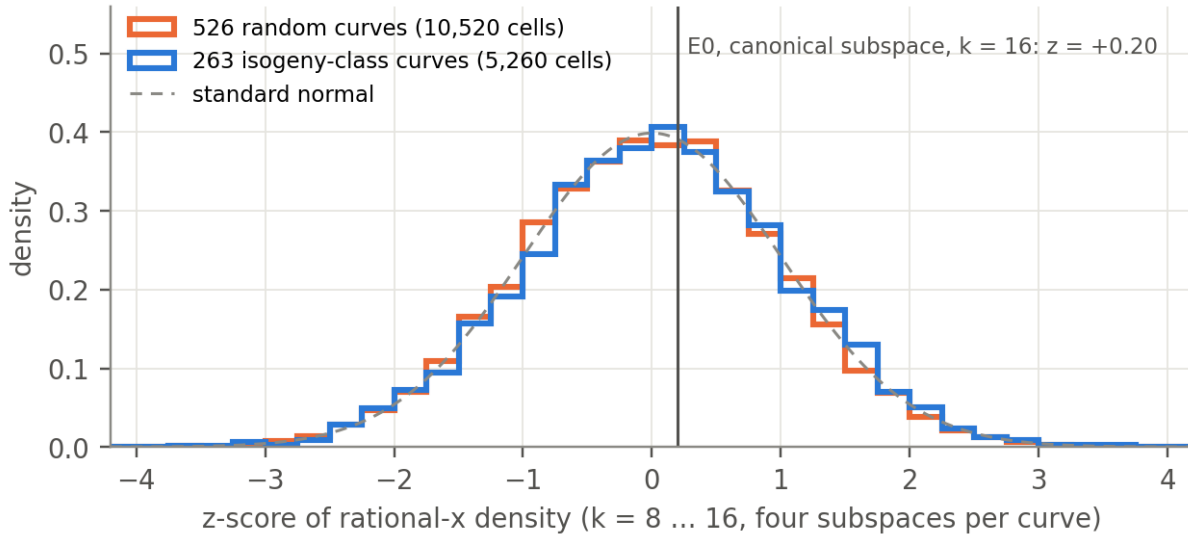


Figure 4. Rational-x density z-scores over 5,260 (curve, subspace, k) cells for the 263 class curves, against 10,520 cells for 526 random, non-isogenous curves over the same field. The two distributions coincide with each other and with a standard normal. The class sample is effectively two curves, one per Frobenius orbit.

- Relation probability $\approx |F|^m / (m! \cdot 2^{131})$ depends on the factor-base size and the rational-lift condition $\text{Tr}(x + b/x^2) = 0$, not on the endomorphism ring.
- Only E0 admits a τ -invariant factor base (Galbraith–Granger–Merz–Petit), worth up to $m!$ in relation collection; that gain carries over to the floor curves through the isogeny.
- Per-curve Semaev point-decomposition problem at $n = 131$ (gap G5): Codex's E0 $d_{\text{reg}} 8$ vs 10–11 is reproduced but is a polynomial-basis presentation effect. Every class curve reaches 7 on $\text{span}\{(b+1)^j\}$, a 0-bit difference after minimising over presentations.
- Cost model (gap G3): no method family beats rho; the tightest non-generic margin is +23.6 bits (Semaev 2015, $D \leq 4$). As for every such claim at $n = 131$, this rests on extrapolation from small parameters.

7. Red team

Three agents generated 44 attack ideas from three angles: generic and endomorphism tricks, algebraic and cover attacks, and structure and edge cases. 37 self-refuted with computations; 4 work only amortised or against flawed implementations; 2 are plausible; 1 is open. The seven that were not self-refuted were each checked by two verifiers.

Idea	Verdicts	Outcome
RT0-6 Kuhn–Struik batch over the 263 curves	partial, confirmed	no single DLP below $2^{60.81}$; 263 instances cost $2^{65.50}$ in total ($2^{57.46}$ each)
RT0-7 Bernstein–Lange precomputation on E0	partial ×2	precomputation $2^{70.8} - 2^{80.8}$ exceeds one rho; batch rho is better for known instance sets
RT0-8 twist / x-only oracle	partial ×2	leaks $k \bmod 2 \cdot 263$ on E0 (not $2 \cdot 263^2$); stopped by an on-curve check; ECDLP unchanged
RT2-13 pooling on E0 across curves	partial ×2	$0.5 \cdot \log_2 L - 0.675$ bits amortised

Idea	Verdicts	Outcome
RT15 / RT2-01 Kani transport of the p-levels	partial ×4	conditional on a characteristic-2 dimension-2/4 algorithm; dimension 8 as proved gives no gain (§5)
RT8 small-genus cover of the 130-dim Weil factor	refuted numbers, partial	break-even genus ≈ 273 –314; cyclic order-131/263 covers excluded below genus 525; no construction known; open

8. Corrections made during verification

All corrections below are already folded into the tables and figures above.

- Level 263p has $262(p + 1) \approx 2^{65.06}$ curves, not about p; p is inert, not split.
- The GHS magic number on the floor is exactly 131, not “130 or 131”.
- The claim that transport from the p-levels is “infeasible by known methods” was refuted; it is conditional (§5).
- The x-only twist leak on E0 is $\pm k \bmod 2 \cdot 263$, not $\bmod 2 \cdot 263^2$. $2^{53.27}$ is E0's twist and $2^{56.79}$ the floor twists'. Subtract 1 bit for the sign.
- Transport evaluation takes 788 additions, not 787. $2^{15.59}$ is the maximum over 262 measurements; a full rebuild is $\approx 2^{16.19}$.
- Batching gains were overstated: $0.5 \cdot \log_2 L - 0.675$ bits with independent base points.
- The p-level construction cost is a best-known $2^{70.0} - 2^{71.3}$, not a $2^{72.97}$ lower bound.
- The isogeny-walk reference to “Menezes-Qu-Teske” in the original prompt was wrong; the relevant work is Galbraith-Hess-Smart and Hess.

9. Relation to the Codex investigation (Runs 01-13)

Codex's numbers are correct (0 mismatches across 263 curves) and its “no advantage from descending” verdict stands. However, its per-descendant signals are artefacts:

- The 262 descendants are two curves up to coordinate Frobenius, so per-descendant differences are factor-base presentation effects.
- Run-08's best ratios ($0.7075 \rightarrow 0.9766$ as k grows) match a pure-chance best-of-(262×64) model. Its “membership degree no worse” gate is a parity coin flip.
- E0's d_{reg} 8 vs 10-11 comes from $b = 1$ being short in the polynomial basis; curves with $b = t$ reach 7.
- B067's +20.7% attempt reduction is reproducible on E0's own subspaces (a selection effect).
- “Zero natural rows” was certain: expected relations were $2^{-96} \dots 2^{-108}$ per probe.
- Runs 09-12 measured quantities fixed by isomorphism invariance; Run-12's 4.07% is unstable (8.8% in Run-13). Run-13 counts targets in the whole group rather than the order-N subgroup (3.68 – $4.08 \times$ overstatement).
- What survives: a clean, verified CM and volcano dataset, including the Run-11 CM scalar alignment ($\delta \equiv 775 - \omega$, $\beta\gamma \equiv -11$ from the reverse map being $-\phi$).

10. Coverage and residual uncertainty

Stage	Planned	Returned	Outcome
Workflow 1: sweep-claim verifiers	120	67	54 confirmed, 12 partially, 1 refuted; 53 lost to session limit
Gap fills G1–G6	6	6	rho endomorphisms and GHS re-verified on all 263; literature and levels reconciled; 48,220 cross-sweep comparisons, 12 errata
Workflow 2: backfill verifiers	22	20	all group-order, endomorphism-ring and pairing claims confirmed except group-order C5 (partially, corrected); 2 audits lost to the weekly limit
Workflow 2: p-level crux	3	3	interval [60.81, 64.33] confirmed; Galbraith 2024 read from source; construction figure corrected
Red-team idea verifiers	14	14	1 confirmed, 12 partially, 1 refuted

- **Not established:** whether a characteristic-2 dimension-2/4 Kani isogeny pipeline can be built. That decides where in [60.81, 64.33] the p-levels sit. The small-genus cover question (RT8) is also open. Neither affects any curve that anyone can write down today.
- Two audit verifiers (group-order C4, C5) never ran; both claims have confirming or correcting recomputes.
- Index-calculus statements extrapolate from small parameters; the margin to rho is ≥ 23.6 bits.

11. Files and reproduction

All paths are relative to /Volumes/SSD990/ecdlp-hardness-work/.

Path	Contents
report/per_curve_hardness.csv	263 rows × 19 columns, 0 empty cells (Appendix A)
report/REPORT.md, report/claims_ledger.json	Markdown report; every claim with verdicts and corrections
ground_truth/, ground_truth_check/	curve build (≈ 61 s: sage -python build_ground_truth.py), loader, independent Vélu derivation
group-order/ ... literature/ (12 sweeps)	scripts, raw outputs, per_curve.json per attack vector
verify/, verify2/, gaps/G1...G6/, redteam-*/	verifier, gap-fill and red-team scripts and results
codex-review/REVIEW.md	run-by-run review of the Codex investigation
report/pdf/	this PDF's figure and layout scripts

Appendix A. Per-curve hardness table (all 263 explicit curves)

j in hex (bit i = z^i of the ECC2K-130 polynomial basis). All curves use $a_2 = 0$ and have $\#E = 4N$, $E(F_q) \cong Z/4N$, embedding degree 118 bits. Native and effective costs are $\log_2$ rho iterations. Transport is $\log_2 F_q$ -multiplications to build the 263-isogeny and evaluate two points. m is the GHS magic number. IC z is the canonical-subspace density z-score at $k = 16$.

Curve	Level	j (hex)	m	Class	Native	Transp.	Effective	IC z
E0	1	0x1	1	262	60.81	—	60.81	+0.20
A000	263	0x1e21822785c5ff89ee7db2942f20ee0	131	2	64.33	15.54	60.81	-0.09
A001	263	0x44049193d18d17979d191b975bd40158	131	2	64.33	15.54	60.81	+1.37
A002	263	0x50f5f404b3ca18cb242a4cd22b5e90996	131	2	64.33	15.54	60.81	-0.11
A003	263	0x3a4747e853e101e2a67576ac0987361f3	131	2	64.33	15.54	60.81	+0.05
A004	263	0x70fcf5c2e4811ac1bf21195e044919bd3	131	2	64.33	15.54	60.81	+1.09
A005	263	0x1607003716fd607f49e2c44f0dba6445a	131	2	64.33	15.55	60.81	-0.99
A006	263	0xac80093004473c31e9b442f9c558479c	131	2	64.33	15.54	60.81	+0.77
A007	263	0x2ef77d111428b621105f4b58335302de	131	2	64.33	15.55	60.81	+0.87
A008	263	0x53e1f06d3917f5a4a5fa13a96b7992da	131	2	64.33	15.55	60.81	+1.06
A009	263	0x45fa20590d4340de2f319a0f8517bcaf2	131	2	64.33	15.54	60.81	-1.40
A010	263	0x70ec04ac2810a8ce467fb562a142398bb	131	2	64.33	15.59	60.81	+0.45
A011	263	0x42560e3040e6fa347bd11f53fb976254c	131	2	64.33	15.54	60.81	+0.00
A012	263	0x3028fedd17492255a1525cdb14ad26657	131	2	64.33	15.55	60.81	+0.64
A013	263	0x77013e97540a397b5bc2e79f7d38ff24d	131	2	64.33	15.54	60.81	+2.08
A014	263	0x283411e644adfc3bbe156ed259ac5e8e	131	2	64.33	15.54	60.81	-1.41
A015	263	0x446f8427b11e4d4c9de5e3ab384170302	131	2	64.33	15.54	60.81	-0.75
A016	263	0x70e0ca44c1b12dc5a25a095df76ab2e03	131	2	64.33	15.54	60.81	+1.17
A017	263	0x342585263ade1f6601cb35e3f3971b5a	131	2	64.33	15.56	60.81	-3.23
A018	263	0x437f16cb990fe79dc71ba7c5d5c07cc4	131	2	64.33	15.55	60.81	-0.05
A019	263	0x14a43381475b9f32f0dc8e56515d948fe	131	2	64.33	15.55	60.81	+0.10
A020	263	0x4fe2aca56374d54ab496a8f4dd123755a	131	2	64.33	15.55	60.81	-1.39
A021	263	0x6750586bee62acea8862d2e77bbeb722d	131	2	64.33	15.58	60.81	-1.52
A022	263	0x1897aa15f8d4cff10a0d9c6bff361e9d8	131	2	64.33	15.54	60.81	+0.38
A023	263	0x1986dad0aefdl1fe814972bb62350432c0	131	2	64.33	15.55	60.81	+0.62
A024	263	0x4c8c85da3daf2e56ee5e053a58145eb60	131	2	64.33	15.55	60.81	-0.67
A025	263	0x36368627d5ac38ad7078eb422a49fa169	131	2	64.33	15.54	60.81	+0.33
A026	263	0x679c72dbc58c6cd17bb8be2570d2faf19	131	2	64.33	15.54	60.81	-0.47
A027	263	0x5cd5dc25972a78a060fc96319b726f79e	131	2	64.33	15.54	60.81	-1.70
A028	263	0x39226fbd7d2438bcd966ea981c3b9a05	131	2	64.33	15.54	60.81	+1.65
A029	263	0x60db2c6ccf45b42e947f39e470ca61b47	131	2	64.33	15.54	60.81	-0.46
A030	263	0x5d5379f26a618e777757b0edf7f57b3aa	131	2	64.33	15.54	60.81	-0.97
A031	263	0x787eba32d0c9163c8fedd909d25c97275	131	2	64.33	15.55	60.81	-0.40
A032	263	0x5580811fa0e9266e6a5236cf1ec92564e	131	2	64.33	15.54	60.81	-0.76
A033	263	0x7aac8043911e5905b8561e9b7ccd8fe65	131	2	64.33	15.55	60.81	+1.16
A034	263	0x10aeba239145a4249b5e0c1ca8ca54fa0	131	2	64.33	15.55	60.81	+2.35
A035	263	0xb47bf4ab1b97d718e0fb5a8a8e07c9b8	131	2	64.33	15.55	60.81	+1.18
A036	263	0x42f3b4d4a053b6ffa358b0dbc52ad6576	131	2	64.33	15.54	60.81	-0.72
A037	263	0x352b1375f2a26bb6bd51fb35b2df11845	131	2	64.33	15.54	60.81	-0.69
A038	263	0x22ee6be21f22d2b8177fa8799aeb1b7011	131	2	64.33	15.55	60.81	+0.93
A039	263	0x3c7f0b1dcab50c7958126c6e2e3fb170f	131	2	64.33	15.58	60.81	-0.38
A040	263	0x2435850606e31c3c37bfa3ae5624915e3	131	2	64.33	15.57	60.81	+1.40
A041	263	0x68f033bd80bf459c049a59661c2130c3d	131	2	64.33	15.54	60.81	+0.35
A042	263	0x4fc743007370a42d8456b142e0a8cc5b6	131	2	64.33	15.55	60.81	+0.52
A043	263	0x36513593f0270504d3728302bce52ab1d	131	2	64.33	15.54	60.81	-0.29
A044	263	0x22c9ff0e66fcdb31020039c204ab502d1	131	2	64.33	15.55	60.81	-0.26
A045	263	0x2c6b36d84156061dcc8b05633cf237feb	131	2	64.33	15.54	60.81	-0.06
A046	263	0x3b759ea3239a29d0f14ebe9d51dd5cc5f	131	2	64.33	15.54	60.81	-1.22
A047	263	0x20e383fcc91eec26787199b97acd3fc1d	131	2	64.33	15.55	60.81	-1.34
A048	263	0x79035820c1751562779e1d5f0f343c2df	131	2	64.33	15.54	60.81	+0.24
A049	263	0x44db5126f9913953201ac7aa1efad87bc	131	2	64.33	15.55	60.81	-0.41
A050	263	0x34b23db3ae5f69cb10889b1b65deeb0f	131	2	64.33	15.55	60.81	-0.50
A051	263	0x37a6b368707cf371fe18bd03cbcc7368d	131	2	64.33	15.55	60.81	-0.25
A052	263	0x3290bd9ea636de95d701c523432d2d31	131	2	64.33	15.55	60.81	+1.18
A053	263	0x372b9f42e428dc0fb9cd62cb6d5fce659	131	2	64.33	15.55	60.81	+0.70
A054	263	0x23d26ef18912f16e1ce13a826831069af	131	2	64.33	15.54	60.81	+0.83
A055	263	0x79712b3d9a5f065c87cf4b6b125bcbfdb	131	2	64.33	15.57	60.81	+1.88
A056	263	0x119b924e7eb3216c99acb90600cce921a	131	2	64.33	15.54	60.81	+2.79
A057	263	0x4a4898f0de3de74dc41ff233390b8b4fc	131	2	64.33	15.54	60.81	-0.88
A058	263	0x66a8e686c8831206381c643c1883f7b6f	131	2	64.33	15.54	60.81	+0.71
A059	263	0x1dcbbb8709dec1cb6786c0225f7d44fe4	131	2	64.33	15.54	60.81	-0.16
A060	263	0x182833a1e5b0808742892c4a351822c26	131	2	64.33	15.54	60.81	+1.10
A061	263	0xcd43ed12234e94958bdb203d04284a1a	131	2	64.33	15.56	60.81	+0.48

Curve	Level	j (hex)	m	Class	Native	Transp.	Effective	IC z
A062	263	0x3233aa1715b2f5a297d23b71e9ab7972	131	2	64.33	15.54	60.81	-0.61
A063	263	0x5262c7764efac1e871bb8e72e53f8e8a	131	2	64.33	15.54	60.81	-0.52
A064	263	0x40bb2d986a20d28288b71a14ef30495c4	131	2	64.33	15.54	60.81	-0.18
A065	263	0x7556a5f27f7d99949fe87c437a3b84baf	131	2	64.33	15.54	60.81	+0.31
A066	263	0x7febbdeb8ad533cc442d169dcd8dbfd2	131	2	64.33	15.55	60.81	+0.85
A067	263	0x5481444ff1f1381eb192eb2abb3760704	131	2	64.33	15.55	60.81	+2.30
A068	263	0x2ef7d44fa0a5f251571d460f75c55ce17	131	2	64.33	15.54	60.81	+0.74
A069	263	0x3f4e17806bb5b704f812ecbd4779bef2d	131	2	64.33	15.55	60.81	+1.00
A070	263	0x2543f3581bd1d0008ba0b3bca39210731	131	2	64.33	15.55	60.81	-0.65
A071	263	0x78bbf5704232b8d519a12a0d7e0550b01	131	2	64.33	15.55	60.81	-0.84
A072	263	0x50c6a0e157eed710b439267ab1d63fb86	131	2	64.33	15.54	60.81	+0.05
A073	263	0x2b0625cbec42480febd55e5e7bd34bbab	131	2	64.33	15.54	60.81	+0.10
A074	263	0x3fe610cd79bd23e00b34f479e7c547ffd	131	2	64.33	15.55	60.81	-1.55
A075	263	0x6400099d4b16722bc58cf57e6c6559807	131	2	64.33	15.55	60.81	+0.86
A076	263	0x59f4051502c89c3ee6f24a88c3297ed7c	131	2	64.33	15.56	60.81	-0.20
A077	263	0x798c06a544ee4aea147b37c6d96171b81	131	2	64.33	15.54	60.81	+0.38
A078	263	0x1498d735408afdafe1e71f58a6ab4eeef	131	2	64.33	15.55	60.81	-1.07
A079	263	0x5ea6d8963ac366eab156d2be013cba2fa	131	2	64.33	15.54	60.81	+1.12
A080	263	0x2c4fbdc6d82dac921aa7037999b5b5ad	131	2	64.33	15.54	60.81	-0.38
A081	263	0x7f61f640b17837189f78803b6dfe67269	131	2	64.33	15.54	60.81	+0.02
A082	263	0x507d94c52d8a512b64b818437c309928	131	2	64.33	15.54	60.81	+0.09
A083	263	0x11aa50d1298ee621ab3e7cd7fcbbl7c0	131	2	64.33	15.55	60.81	-1.75
A084	263	0x1b4ceab9af463e4b3e9cc20e4c983d358	131	2	64.33	15.54	60.81	-0.71
A085	263	0x1ce6e3774aabe1d9fa57f58d6458b694e	131	2	64.33	15.55	60.81	+0.59
A086	263	0x4c275c8f4c6797d6b3abfab15d1878bda	131	2	64.33	15.55	60.81	-0.09
A087	263	0x323138d7a9329753f3759c918e847a423	131	2	64.33	15.54	60.81	+0.13
A088	263	0x6729264a34c22fd1ff8b277f9ed3beadd	131	2	64.33	15.54	60.81	+0.66
A089	263	0x4c927edb7d98b7a837a771793a456a7b6	131	2	64.33	15.55	60.81	-0.03
A090	263	0x2276ca3c911b7cf2d17c9dd7a394be5cb	131	2	64.33	15.54	60.81	-0.31
A091	263	0x2d6c97ce77f934778e0b59e6956753dfd	131	2	64.33	15.55	60.81	+0.33
A092	263	0x7b3acf77ced527599294559b23f4c36d1	131	2	64.33	15.54	60.81	-0.73
A093	263	0x4a330af37079690374a7eee7cc2bdcb0	131	2	64.33	15.55	60.81	+0.19
A094	263	0x40b3fd377253bf5223d650d4cb8e9a538	131	2	64.33	15.54	60.81	+0.38
A095	263	0x3117f220042e6292d62dba24e546ffec7	131	2	64.33	15.58	60.81	-1.08
A096	263	0x670b1e800378790140e43f1c9d2adfe23	131	2	64.33	15.54	60.81	+0.53
A097	263	0x18c202e3480b95055076949580cf112da	131	2	64.33	15.54	60.81	-0.50
A098	263	0x983317d043e0d32e39456a1defb41cca	131	2	64.33	15.55	60.81	-0.34
A099	263	0x13d8d077621ed57f14f0611d423a67312	131	2	64.33	15.56	60.81	+0.47
A100	263	0x4b6029c22b5586c6d82d4d5e6aeac17ea	131	2	64.33	15.55	60.81	-0.87
A101	263	0x32b69c016a9d74683bb7faad94df6411b	131	2	64.33	15.54	60.81	-0.72
A102	263	0x732af7dad924000bdaeadd7b92f5cf4cb	131	2	64.33	15.55	60.81	+1.46
A103	263	0x3626abb17952dbf3ccf69f5055557e9a	131	2	64.33	15.55	60.81	+0.16
A104	263	0x1032d8788effa4e35f80deeb7b6db542a	131	2	64.33	15.56	60.81	+2.01
A105	263	0x1a15226e7d9581c7c25f4145fcf21c4aa	131	2	64.33	15.54	60.81	+1.37
A106	263	0xdbd5abe3d3cdf19d02c8eadc10874a7c	131	2	64.33	15.54	60.81	+0.98
A107	263	0x4239b43bf9aaf119742a25377c0248d5e	131	2	64.33	15.54	60.81	+0.95
A108	263	0x30682089f6a131b4519caeba124b5d1b3	131	2	64.33	15.55	60.81	-0.54
A109	263	0x2304cf846d578288579afd1f725ca668b	131	2	64.33	15.54	60.81	+0.22
A110	263	0x69600b3664190a5cf6aab0d16ec7645	131	2	64.33	15.54	60.81	+0.67
A111	263	0x1bda9c9c5517e336ccc9f5089e42ec820	131	2	64.33	15.54	60.81	+2.08
A112	263	0x55ce179facc278d20bbea5b915c7666ad8	131	2	64.33	15.57	60.81	-0.52
A113	263	0x2c36591f948851be77643980833b3ca27	131	2	64.33	15.54	60.81	-0.39
A114	263	0x6b747388f9cf5d159ea3d390ea58a1f43	131	2	64.33	15.56	60.81	-1.16
A115	263	0xbb693f4967186d645d93306ed2b4a834	131	2	64.33	15.56	60.81	-0.98
A116	263	0x53e5b28a8b2147e3ce88c25990da2e870	131	2	64.33	15.55	60.81	+0.62
A117	263	0x3e750da9a66b87bed2b3684bf08108ddf	131	2	64.33	15.55	60.81	-0.71
A118	263	0x2058c6ab167ceeddcd8ba0d580e09b5	131	2	64.33	15.54	60.81	-0.30
A119	263	0x7945fc92759ebbf61a98c0730f0f2ddc7	131	2	64.33	15.54	60.81	+0.23
A120	263	0x5cea1bc4523db2c83e8995aa5f0028f2	131	2	64.33	15.54	60.81	+1.52
A121	263	0x54ad365f6fd48a520e0fa3c642ce63e304	131	2	64.33	15.56	60.81	+0.48
A122	263	0x3ff2ae6b678cf901547bb3c7acbf6db99	131	2	64.33	15.54	60.81	+0.05
A123	263	0x7115137aee1c9fe7cdc5c362509f44cf9	131	2	64.33	15.54	60.81	-0.85
A124	263	0x135f4fba4b3680ef4b080867d94008d90	131	2	64.33	15.55	60.81	+0.08
A125	263	0x4e33ad43b605f0fcb6e226db91120f7e0	131	2	64.33	15.55	60.81	-0.76
A126	263	0x335c2761be2aa524a6cc45ed461645ae9	131	2	64.33	15.55	60.81	+0.25
A127	263	0x3327e9647cc3cc49b5093bfe8f8b697f7	131	2	64.33	15.55	60.81	+1.99
A128	263	0x262369d54adf88f1956760702364e2043	131	2	64.33	15.56	60.81	-0.75
A129	263	0x28c83c77cfc87affb7bf7c40d68713eeb	131	2	64.33	15.54	60.81	+0.04
A130	263	0x7ec73785652482c526016781d0bf00ffd	131	2	64.33	15.55	60.81	-0.36
B000	263	0xb0e0455e0f58166e08e8e6f43c15655	131	2	64.33	15.55	60.81	+0.02
B001	263	0x5403a7061801e5febcd4d1206fcd44bc9f	131	2	64.33	15.56	60.81	+0.35

Curve	Level	j (hex)	m	Class	Native	Transp.	Effective	IC z
B002	263	0x3be05420e89301cd5885440809fc50f52	131	2	64.33	15.54	60.81	-0.23
B003	263	0x74b15d55ee029134fd79b37441932a84	131	2	64.33	15.55	60.81	+0.91
B004	263	0x42f6e30ebd7aeaa94f5b4d002b9e1f6c1	131	2	64.33	15.58	60.81	+0.73
B005	263	0x702f578b1d33120fb46bd5ea5feabf0e6	131	2	64.33	15.55	60.81	+0.01
B006	263	0x6143b46afd187a3597a32335b68574cb	131	2	64.33	15.55	60.81	+1.77
B007	263	0x148d5bf121f3a19ffb447a09c085ed31d	131	2	64.33	15.54	60.81	+1.04
B008	263	0x4aa6c63eeeb5131b6d480620ad49997bf	131	2	64.33	15.54	60.81	-0.55
B009	263	0x37afa8de60cd604b0f36f2bb5a66586d2	131	2	64.33	15.55	60.81	+0.34
B010	263	0x6290bb00ff923c09d856672fc815f27d2	131	2	64.33	15.54	60.81	-0.66
B011	263	0x5c7a8e13b4e6500301149e79652507fdb	131	2	64.33	15.55	60.81	+0.60
B012	263	0x6821d4aa96580b60e2f61d88ef454497a	131	2	64.33	15.54	60.81	+1.61
B013	263	0x1ac53d096ae6affbfcfd9ec8047a69c395	131	2	64.33	15.55	60.81	+0.79
B014	263	0x1cea35af307a469ecfea088afb0511929	131	2	64.33	15.57	60.81	+0.26
B015	263	0x1d271ebc63fcfe536701d0ade09bd92af	131	2	64.33	15.55	60.81	+0.75
B016	263	0xd3b78960c0bf574cd6554634508be463	131	2	64.33	15.55	60.81	+0.20
B017	263	0x472e31f7d4c7ce9d026152f546eb93aeb	131	2	64.33	15.55	60.81	-0.25
B018	263	0x21827a09730d53807ba37094634b9da94	131	2	64.33	15.55	60.81	-1.07
B019	263	0x3c58c53885ac57ced673437bab25c0f1e	131	2	64.33	15.55	60.81	-2.07
B020	263	0x2071a9d621ef71d395a9fe30bac0779ba	131	2	64.33	15.55	60.81	+1.59
B021	263	0x2c54c308ba883fdc395d1e17d31c2a81c	131	2	64.33	15.55	60.81	-0.17
B022	263	0x6b35efff235660387b5fb97d525413d988	131	2	64.33	15.55	60.81	+0.38
B023	263	0x4be737ac1f41466c32fad8c9053081bf1	131	2	64.33	15.54	60.81	-1.45
B024	263	0x66f5b4ed673c4bf254846f40fe5c8cbe8d	131	2	64.33	15.58	60.81	+1.91
B025	263	0x499e13bfe6286ecc32475b3f0123cc9f	131	2	64.33	15.54	60.81	+0.52
B026	263	0x3388cc180e74b00509c82ed59a5a32f84	131	2	64.33	15.55	60.81	+0.05
B027	263	0x6230939726751c81125dd3e3e6441037e	131	2	64.33	15.55	60.81	+1.97
B028	263	0x49393357ce258f867cdd97ba0d77943bd	131	2	64.33	15.54	60.81	-1.59
B029	263	0x77db61db6327ee80674d39acc24091038	131	2	64.33	15.55	60.81	+1.03
B030	263	0x43d52da6d8486db299726dc45b5e910ff	131	2	64.33	15.55	60.81	+0.33
B031	263	0x20607fbb2f7dad8a7dc1f3d09a2ca6ee4	131	2	64.33	15.55	60.81	+0.44
B032	263	0x6815dd40d451e4a76a42c17daa3dc219e	131	2	64.33	15.54	60.81	-0.55
B033	263	0x1f855a9cf297fa1532ee0ee1c49b2a9add	131	2	64.33	15.55	60.81	-0.42
B034	263	0x181484babdaf41717acf411b85abf167	131	2	64.33	15.55	60.81	-0.87
B035	263	0x59811bb280b2b4fbb079eb12f01fc379b	131	2	64.33	15.54	60.81	+0.55
B036	263	0x6d8c944a18a4b67ec142b22aa3f21017a	131	2	64.33	15.56	60.81	-1.62
B037	263	0xb689363dab1a7ac5cd2f9e6a3998162d	131	2	64.33	15.54	60.81	-0.43
B038	263	0x53a3db879f22cd203caff31ce9d0bb2b1	131	2	64.33	15.54	60.81	+1.52
B039	263	0x6a30a9316de0c58588423c2b762f79e88	131	2	64.33	15.55	60.81	-0.23
B040	263	0x5ef92753afce374bdf2d095664e8d1871	131	2	64.33	15.55	60.81	-1.16
B041	263	0x6d5b519b7886ae61ae45261862c2e3506	131	2	64.33	15.54	60.81	+0.05
B042	263	0x5a7ea8e6ff1cdce785c3918bfe1dc9613	131	2	64.33	15.54	60.81	+0.27
B043	263	0x69f8c41fbe9249db40420c227c660d734	131	2	64.33	15.55	60.81	+0.19
B044	263	0xedc41c765b51c41f01bcb68b2d938899	131	2	64.33	15.57	60.81	-0.73
B045	263	0x4707fd35b05961d6d9ece2aa5d64e4141	131	2	64.33	15.54	60.81	+1.34
B046	263	0x34d60307d46e23eca741e98bcd587e586	131	2	64.33	15.56	60.81	-0.66
B047	263	0x63f76ecc042750c46ee4ce5838654fb94	131	2	64.33	15.55	60.81	-0.60
B048	263	0x483552c6ca0e776455e43d87ff4735917	131	2	64.33	15.57	60.81	+1.17
B049	263	0x73d133afaf2a759f36fc074dc2d68d7fc	131	2	64.33	15.54	60.81	-0.02
B050	263	0x53307a4ff6671ef55fa022fb883bc9d5b9	131	2	64.33	15.54	60.81	-0.45
B051	263	0x7b36331194f9f257ddccd61c0a12ab01e	131	2	64.33	15.54	60.81	-1.61
B052	263	0x1e6329c67364b499ff310881638ff96b	131	2	64.33	15.55	60.81	-0.21
B053	263	0x405d09c9366b8c6588d366ccf2f2a2c9d	131	2	64.33	15.55	60.81	-0.03
B054	263	0x6051bd2f16d827ce638cdbe5c6318d78e	131	2	64.33	15.55	60.81	-0.11
B055	263	0x1c14ba5ce02e3a061bcb3c9169c5323dd	131	2	64.33	15.55	60.81	+0.99
B056	263	0x19711be2e4e8e8721d003b1965fc05f89	131	2	64.33	15.55	60.81	-0.12
B057	263	0x5dde965f0ce4592e0ce61b935fd5c20a1	131	2	64.33	15.54	60.81	+2.02
B058	263	0x3c3d3c5e9ed9bd1303324da5acca6eade	131	2	64.33	15.54	60.81	-1.26
B059	263	0x3464247b3421bc1adc0e3898b06e5bfb6c	131	2	64.33	15.55	60.81	-0.52
B060	263	0x26e4dca469b580f667e12c18a1615825e	131	2	64.33	15.55	60.81	+0.38
B061	263	0x6c8a0ca73967a9f4cad8a6fc8142072ec	131	2	64.33	15.55	60.81	+0.78
B062	263	0x4f3797a85323ac82709a8985ef8d56cb9	131	2	64.33	15.55	60.81	+0.63
B063	263	0x721772d4dbd1bed4ea636c30ba37d5e1e	131	2	64.33	15.54	60.81	+3.33
B064	263	0x123c0bc6877f2ab66db56e0db01f3faeb	131	2	64.33	15.54	60.81	-0.64
B065	263	0x4a78646012f075ca958381177fabf3a45	131	2	64.33	15.55	60.81	+0.54
B066	263	0x77fcd0d0ddfc9104e2f1711d2b877340	131	2	64.33	15.54	60.81	-0.46
B067	263	0x6c104736b136a4228691612728b4e5bf	131	2	64.33	15.54	60.81	-2.09
B068	263	0x50da641a14e18366cbb34f76daf8e06e3	131	2	64.33	15.54	60.81	+1.30
B069	263	0x7e0638edde05cb90fa65c09852388234	131	2	64.33	15.54	60.81	-1.01
B070	263	0x500940c935970b2d397d1c00f826e08a1	131	2	64.33	15.54	60.81	-0.70
B071	263	0x3a01129ee11322cf32a88f8202b77e966	131	2	64.33	15.55	60.81	-0.10
B072	263	0x65fc044b1b3a981b180b1e66796f76c1a	131	2	64.33	15.55	60.81	+1.49

Curve	Level	j (hex)	m	Class	Native	Transp.	Effective	IC z
B073	263	0x19b9513411b0e2a618f660badc805cc1b	131	2	64.33	15.55	60.81	-1.95
B074	263	0x48d8a08fef4e62e54a49a3035c3af8fa5	131	2	64.33	15.58	60.81	-0.13
B075	263	0x73927897e846d6110a43b21c992a09218	131	2	64.33	15.55	60.81	-0.37
B076	263	0x2358b3c9586cec14b95b0de6a9d0b429	131	2	64.33	15.56	60.81	-1.06
B077	263	0x52877e892e730669aac81de730a204a1	131	2	64.33	15.54	60.81	+0.55
B078	263	0x55ba7bc183d05e83cb6e1dd67304dc981	131	2	64.33	15.54	60.81	-0.19
B079	263	0x3bedf5a994a0214d84256b50e84fd18e8	131	2	64.33	15.54	60.81	-0.35
B080	263	0x61a14f3843acfbccdbe2bc556c0bf2ff6	131	2	64.33	15.54	60.81	-0.36
B081	263	0x1d58bc5ff7072090f421aa6751f023525	131	2	64.33	15.54	60.81	-0.35
B082	263	0x183ebcc3f574f94106c611d66813da77f	131	2	64.33	15.55	60.81	+0.42
B083	263	0xd90254fb177243057aad3e605ceb16d5	131	2	64.33	15.57	60.81	+0.35
B084	263	0x4279cb416caef355f25e929669b1271c7	131	2	64.33	15.55	60.81	+0.35
B085	263	0x352994d837f2b45b8e62fe0a5332aa6ca	131	2	64.33	15.54	60.81	+0.25
B086	263	0x72bf2b8ccff3688473c4173bf6f5e86a4	131	2	64.33	15.56	60.81	+0.83
B087	263	0x173bf5477af1c323675142338f93c5f97	131	2	64.33	15.54	60.81	+0.39
B088	263	0xed670e012abf5d3c5af125920233c1cd	131	2	64.33	15.55	60.81	-1.01
B089	263	0x160a2ec8d3424c515f2aeea157ac7b3d1	131	2	64.33	15.54	60.81	+0.27
B090	263	0x1b9946e96a0f37d33b63a12db371eb259	131	2	64.33	15.55	60.81	+0.88
B091	263	0x9b489dfb5db5b8b8bfc120626c52b14f	131	2	64.33	15.55	60.81	-0.35
B092	263	0x569cf3f3c7983d51a3f928f7f9f482bed	131	2	64.33	15.54	60.81	+0.39
B093	263	0x3edacc32477153656490ddd0717bfef38	131	2	64.33	15.54	60.81	+1.02
B094	263	0x600e69aa7734223cf587c61afe72986f1	131	2	64.33	15.55	60.81	+0.25
B095	263	0x901071d8bdcaab8c2d367a52d7965578	131	2	64.33	15.54	60.81	+1.08
B096	263	0x43cb505a01d61c6c92f02863ebfaf0dd7	131	2	64.33	15.55	60.81	+0.09
B097	263	0x246177f6ae50aded101d3ac95861dbf24	131	2	64.33	15.54	60.81	-0.62
B098	263	0x2de0885bd7d543dfab19bf02fa285bcc8	131	2	64.33	15.54	60.81	-1.59
B099	263	0x2a3d5c07d281b9b56ed1e6eea03168bc0	131	2	64.33	15.55	60.81	-0.44
B100	263	0x7fa9612ba82011d03e3b814ca0c9e3000	131	2	64.33	15.56	60.81	+0.16
B101	263	0x5105aada990e6fa1fa853c551a083b551	131	2	64.33	15.54	60.81	+0.80
B102	263	0x7a4e50ecfaeb6cfbd9db024dde15fca50	131	2	64.33	15.54	60.81	-2.11
B103	263	0x45b8f348fb564b2755ea0fb79b2305f3f	131	2	64.33	15.55	60.81	-0.59
B104	263	0x20a8e4821772a78605a3aa99a3b63d004	131	2	64.33	15.54	60.81	+1.54
B105	263	0x2807eed30ca694690bc3967ebdc480390	131	2	64.33	15.55	60.81	-1.81
B106	263	0x3fd055945e5b7b664773a99b8bf48f700	131	2	64.33	15.58	60.81	+1.30
B107	263	0x60443a10fabd9ee1ed5cf9e395e7d80d8	131	2	64.33	15.55	60.81	-1.37
B108	263	0x1915f4b461ed5b0655faa46d8c4f08791	131	2	64.33	15.54	60.81	-0.46
B109	263	0x5cdff0fed06a6f6f488181a6fa9b310fef	131	2	64.33	15.54	60.81	-1.49
B110	263	0x7d36394342005e3f55cafdce538b42132	131	2	64.33	15.54	60.81	+1.98
B111	263	0x53e33cd249aa176f06c0004ec5830535	131	2	64.33	15.55	60.81	-0.11
B112	263	0x11ba71196221236f6db388bbad4ac289f	131	2	64.33	15.55	60.81	-1.14
B113	263	0x1a48e0ac925a1cdb9c297d4a7d669b7bb	131	2	64.33	15.55	60.81	+2.40
B114	263	0x1ca8b2921c06af63df78aca80d622dd73	131	2	64.33	15.54	60.81	+0.68
B115	263	0x922fec6e37eca6c0874518ebb43cd50b	131	2	64.33	15.57	60.81	+0.71
B116	263	0x468e7c6f514a709e4932c11f63bd8b6a5	131	2	64.33	15.55	60.81	+0.08
B117	263	0x74ded649d0658a47ea4bb3ecc217d89ae	131	2	64.33	15.54	60.81	+0.85
B118	263	0x6f77d4a3a9df2c86b4499a993acb1cbd	131	2	64.33	15.55	60.81	+1.15
B119	263	0x51cf1286856ba6fd35bb9d28123e77d69	131	2	64.33	15.54	60.81	+0.83
B120	263	0x6b0c77430b2ed4fd5bfdf1fccb05926	131	2	64.33	15.54	60.81	-0.60
B121	263	0xab653708782f03217a62b4f7bb66c7ab	131	2	64.33	15.54	60.81	-0.04
B122	263	0x3aaf3d9ef27c3578480c43a12402672b	131	2	64.33	15.54	60.81	-0.44
B123	263	0x5020aef03656cd80a126cd46532ea913	131	2	64.33	15.54	60.81	-0.05
B124	263	0x6b403957eb0b04560422cdbc8b667c282	131	2	64.33	15.54	60.81	+0.06
B125	263	0x1eb6a11574dfee901ff602f351b1efb6d	131	2	64.33	15.55	60.81	+1.20
B126	263	0x94bf69bf80f1bfbd2a268550e4e8b707	131	2	64.33	15.54	60.81	-1.23
B127	263	0x57dbf6b1028bc9f550d446435cb511ba3	131	2	64.33	15.54	60.81	+1.46
B128	263	0x2a8129f146daf65e052e86b526cc5315a	131	2	64.33	15.55	60.81	+1.23
B129	263	0x2aab845a6f89431bb5cf7dff42d9b8a2a	131	2	64.33	15.55	60.81	-1.03
B130	263	0x7bbf6ee0d1a4bdfc8b00d39bf467591f2	131	2	64.33	15.56	60.81	-2.16